The adjacent vertices fault-tolerance for two-spannability of star graphs *†

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Abstract

Let P be a path and $V(P)$ be the set of vertices on P. Two paths P_1 and P_2 are two spanning disjoint paths of $S_n = (V_0 \cup V_1, E)$ if $V(P_1) \cap V(P_2) = \emptyset$ and $V(P_1) \cup V(P_2) = V_0 \cup V_1$. Let F_{av} be the set of f_{av} pairs of adjacent vertices of S_n. In this paper, will show that for any $s_1, s_2 \in V_0$ and $t_1, t_2 \in V_1$, there exists two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of $S_n - F_{av}$ for $f_{av} \leq n-4$ and $n \geq 5$.

1 Introduction

The hypercube Q_n and star graph S_n are the most popular and efficient interconnection networks. They possesses many excellent properties such as recursive structure, symmetry, small diameter, low degree, popular structure embedding, easy routing, and optimal connectivity and fault-tolerance. There are many literatures for fault tolerance of Hamiltonian properties of hypercube and star graphs.

The interconnection network is usually represented by an undirected graph where vertices represent processors and edges represent links. Let $G = (V_0 \cup V_1, E)$ be a *bipartite graph* where V_0 and V_1 are two disjoint vertex sets such that each edge of E consists of one vertex from each set. A *Hamiltonian path* is a spanning path of G. A *Hamiltonian cycle* is a cycle that visits every vertex exactly once. A *Hamiltonian graph* is a graph that contains a Hamiltonian cycle. Let F_e be a set of faulty edges and f_e be the number of F_e. A graph $G = (V, E)$ is k *edges Hamiltonian* if $G - F_e$ is a Hamiltonian graph for every $F_e \subset E$ and $f_e \leq k$. A bipartite graph $G = (V_0 \cup V_1, E)$ is *Hamiltonian laceable* if there exists a *Hamiltonian path* between x, y for any $x \in V_0, y \in V_1$. A bipartite graph $G = (V_0 \cup V_1, E)$ is k *edges Hamiltonian laceable* if $G - F_e$ is Hamiltonian laceable for every $F_e \subset E$ and $f_e \leq k$. The authors of [6] proved that Q_n is $(n-2)$ Hamiltonian laceable with $n \geq 2$. In [3], Li et al. proved that S_n is $(n-3)$ edges Hamiltonian laceable with $n \geq 4$.

In [1], the authors studied the adjacent vertices fault tolerance Hamiltonian laceability of hypercube. Let F_{av} be the set of f_{av} pairs of adjacent vertices. We can denote F_{av} as $\{a_1, b_1, a_2, b_2, \cdots, a_{f_{av}}, b_{f_{av}} | a_i \in V_0, b_i \in V_1$ and $(a_i, b_i) \in E$ for $1 \leq i \leq f_{av}\}$. The authors proved that the hypercube $Q_n - F_{av}$ is Hamiltonian laceable for $f_{av} \leq n-3$ in [1]. In [4], Ou et al. showed that $S_n - F_{av}$ is Hamiltonian laceable for $f_{av} \leq n-3$ for $n \geq 5$.

Let P be a path and $V(P)$ be the set of vertices on P. Two paths P_1 and P_2 are *two spanning disjoint paths* of $G = (V, E)$ if $V(P_1) \cap V(P_2) = \emptyset$ and $V(P_1) \cup V(P_2) = V$. The bipartite graph $G = (V_0 \cup V_1, E)$ is *2-spannable* if for any $s_1, s_2 \in V_0$ and $t_1, t_2 \in V_1$ there exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of G. The authors of [1] showed that $Q_n - F_e$ is 2-spannable for $f_e \leq n-3$. Liu et al. proved that $S_n - F_e$ is 2-spannable for $f_e \leq n-4$ and $n \geq 5$ in [2]. In [5], Su et al. furthermore studied the adjacent fault-tolerance for 2-spannable. The bipartite graph $G = (V_0 \cup V_1, E)$ is *k-adjacent 2-spannable* if $G - F_{av}$ is 2-spannable for $f_{av} \leq k$. The authors of [5] showed that Q_n is $(n-3)$-adjacent 2-spannable for $n \geq 3$. In this paper, we will prove that S_n is $(n-4)$-adjacent 2-spannable for $n \geq 5$.

The rest of this paper is organized as follows. In section 2, we will introduce the definition and

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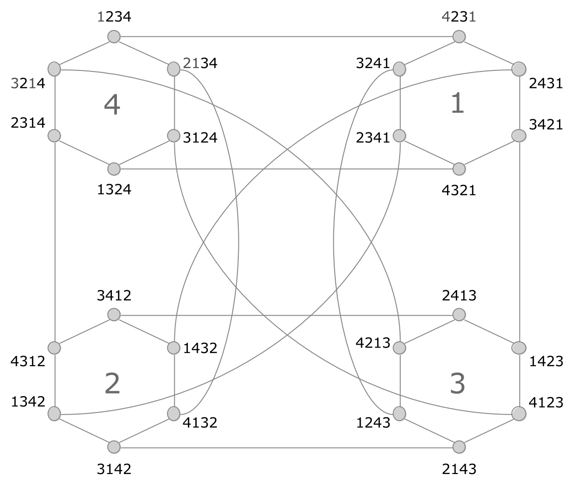

Figure 1: Illustration of S_4.

properties of star graphs. The main result is proved in section 3. Section 4 gives the conclusion.

2 Preliminaries

In this section, we will introduce some definition, notations and properties about star graphs which are useful for our proofs.

Definition 1 *The n-dimensional star graph is denoted by S_n. The vertex set $V = \{v|v$ is a permutation of $1, 2, ..., n\}$ and edge set $E = \{(v_1v_2...v_{i-1}v_i...v_n, v_iv_2...v_{i-1}v_1v_{i+1}...v_n)$ $|v_1...v_n \in V$ and $2 \leq i \leq n\}$.*

The graph S_4 is illustrated in Fig 1.

Let S_n^i be the sub-graph of S_n that the nth digit every vertex is i. Let S_n^I be the sub-graph of S_n induced by the vertices of $\bigcup_{i \in I} V(S_n^i)$ for $I \subset \{1, 2, \cdots, n\}$.

Let $s_1, s_2 \in V_0$ and $t_1, t_2 \in V_1$ be two pairs of vertices. Let $P(s_1, t_1)$ and $P(s_2, t_2)$ be *two spanning disjoint paths* of $G = (V_0 \cup V_1, E)$ if $V(P(s_1, t)) \cap V(P(s_2, t_2)) = \emptyset$ and $V(P(s_1, t)) \cup V(P(s_2, t_2)) = V_0 \cup V_1$. The graph $G = (V_0 \cup V_1, E)$ is *2-spannable* if there exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ for any two pairs of vertices s_1, t_1 and s_2, t_2 with odd distance.

Let F_{av} be a faulty set of vertices with f_{av} pairs of adjacent vertices. The graph $G = (V_0 \cup V_1, E)$ is f-adjacent 2-spannable if $G - F_{av}$ is 2-spannable for $|F_{av}| \leq f$.

A star graph S_n is an edge symmetric and node symmetric graph. By definition, S_n contains $n!$ vertices and $\frac{n!(n-1)}{2}$ edges. The graph S_n can be decomposed to n copies of $(n-1)-$dimensional star graphs. Give $2 \leq k \leq n$, we can use S_n^i to denote the subgraph of S_n induced by the vertices whose nth bit is i. We further denote the induced subgraph $S_n^{i_1} \cup S_n^{i_2} \cdots \cup S_n^{i_k}$ as $S_n^{i_1, i_2, \cdots, i_k}$ for $1 \leq i_1, i_2, \cdots, i_k \leq n, 1 \leq k \leq n$. Let $\langle n \rangle = \{1, 2, \cdots, n\}$ and I_k be the subset of $\langle n \rangle$ with $|I_k| = k$. Let $[u]_k$ be the kth bit of a vertex u.

Li et al. proved the following lemma in [3].

Lemma 1 *The graph S_n is $(n-3)$-edge Hamiltonian laceable with $n \geq 4$.*

Ou and Hung proved the following lemma in [4].

Lemma 2 *Let F_{av} be the set with f_{av} pairs of adjacent vertices. The graph $S_n - F_{av}$ is Hamiltonian laceable for $f_{av} \leq n-3, n \geq 5$.*

3 The adjacent fault-tolerance for 2-spannability of S_n

In this section, we will prove the adjacent fault-tolerance for 2-spannability of star graphs. The following lemma is the base case of the main theorem.

Lemma 3 *The graph S_5 is 1-adjacent 2-spannable.*

Proof: For $f_{av} = 1$, we find that S_5 is 1-adjacent 2-spannable by computer programs. □

Lemma 4 *Suppose that $n \geq 6$. Let F_{av} be the set of faulty vertices with f_{av} adjacent pairs of vertices in S_n^I where every adjacent pair of faulty vertices is in the same subgraph S_n^i for some $i \in I$. If S_{n-1} is $(n-5)$-adjacent 2-spannable, then $S_n^I - F_{av}$ is Hamiltonian laceable for $I \subset \langle n \rangle$ and $1 \leq |I| \leq n$.*

Proof. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, S_{n-1} is $(n-5)$-adjacent Hamiltonian laceable. Thus, this lemma holds for $|I| = 1$. In the follows, we will assume that $k = |I|$ and $I = \{i_1, i_2, \cdots, i_k\}$. Let $S_n^I = (V_0 \cup V_1, E)$ be the subgraph of S_n and s, t be arbitrary vertices of S_n^I for $s \in V_0$ and $t \in V_1$. We will construct a Hamiltonian path $P(s, t)$ of $S_n^I - F_{av}$ for $f_{av} \leq n-5$ in the following cases.

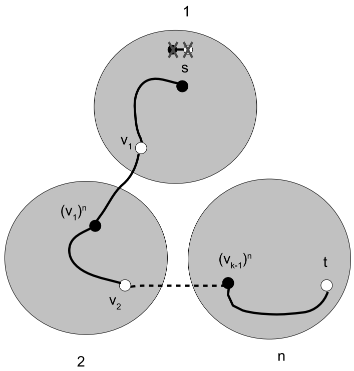

Figure 2: The case 1 for Hamiltonian path of Lemma 4.

Case 1 $s \in S_n^x$ and $t \in S_n^y$ for $x \neq y \in I$.

Without loss of generality, we can assume that $s \in S_n^{i_1}$ and $t \in S_n^{i_k}$. For every $1 \leq j \leq k$, there exists a vertex $v_j \in V_1$ of $S_n^{i_j}$ for $(v_j)^n \in S_n^{i_{j+1}}$ and $v_j, (v_j)^n \notin F_{av}$. Let $(v_0)^n = s$ and $v_k = t$. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, there exists a Hamiltonian path $P((v_{j-1})^n, v_j)$ of $S_n^{i_j} - F_{av}$ for $1 \leq j \leq k$. Thus, the path $\langle s = (v_0)^n \rightarrow P((v_0)^n, v_1) \rightarrow v_1, (v_1)^n \rightarrow P((v_1)^n, v_2) \rightarrow v_2, (v_2)^n \cdots (v_{k-1})^n \rightarrow P((v_{k-1})^n, v_k) \rightarrow v_k = t\rangle$ is a Hamiltonian path between s and t of $S_n^I - F_{av}$, as illustrated in Fig. 2.

Case 2 $s, t \in S_n^x$ for some $x \in I$.

Without loss of generality, we can assume that $s, t \in S_n^{i_1}$. Let $w \in V_1, b \in V_0$ be two vertices of $S_n^{i_1} - F_{av}$ such that $(v)^n \in S_n^{i_2}, (w)^n \in S_n^{i_k}$. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, there exist two spanning disjoint paths $P(s, w)$ and $P(b, t)$ of $S_n^{i_1} - F_{av}$. For every $2 \leq j \leq k$, there exists a vertex $v_j \in V_1$ of $S_n^{i_j}$ for $(v_j)^n \in S_n^{i_{j+1}}$ and $v_j, (v_j)^n \notin F_{av}$. Let $(v_1)^n = (w)^n$ and $v_k = (b)^n$. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, there exists a Hamiltonian path $P((v_{j-1})^n, v_j)$ of $S_n^{i_j} - F_{av}$ for $2 \leq j \leq k$. Thus, the path $\langle s \rightarrow P(s, w) \rightarrow w, (w)^n = (v_1)^n \rightarrow P((v_1)^n, v_2) \rightarrow v_2, (v_2)^n \cdots (v_{k-1})^n \rightarrow P((v_{k-1})^n, v_k) \rightarrow v_k = (b)^n, b \rightarrow P(b, t) \rightarrow t\rangle$ is a Hamiltonian path between s and t of $S_n^I - F_{av}$. □

Lemma 5 *Suppose that $n \geq 6$ and $U = \{s_1, s_2, t_1, t_2\}$ with $s_1, s_2 \in V_0$ and $t_1, t_2 \in V_1$. Let F_{av} be the set of faulty vertices with f_{av} adjacent pairs of vertices in S_n^I where every adjacent pair of faulty vertices is in the same subgraph S_n^i for some $i \in I$. If S_{n-1} is $(n-5)$-adjacent 2-spannable, then there exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of $S_n^I - F_{av}$ with $|U \cap V(S_n^i)| \leq 3$ for $i \in I$ and $2 \leq |I| \leq n$, $f_{av} \leq n - 5$.*

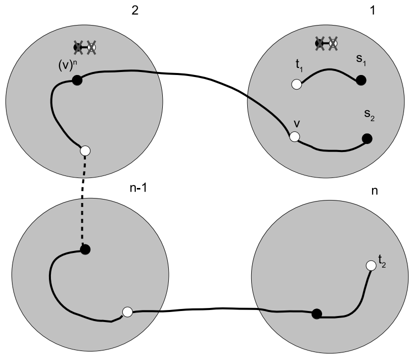

Figure 3: The case 1 for two spanning disjoint paths of Lemma 5.

Proof. Let$U^i = U \cap V(S_n^i)$ for $i \in I$. Let $|I| = k$ and $I = \{i_1, i_2, \cdots, i_k\}$. We will construct the two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of $S_n^I - F_{av}$ with the following some lemmas.

Case 1 $|U^i| = 3$ for some $i \in I$.

Without loss of generality, we can assume that $s_1, s_2, t_1 \in S_n^{i_1}$. Let $J = I - \{i_1\}$. Let $v \in V_1$ be a vertex of $S_n^{i_1} - F_{av} - U^{i_1}$ such that $(v)^n \in S_n^k - F_{av}$ for some $k \in I$. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, there exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, v)$ of $S_n^{i_1} - F_{av}$. By Lemma 4, there exists a Hamiltonian path $P((v)^n, t_2)$ of $S_n^J - F_{av}$. Thus, $P(s_1, t_1)$ and $\langle s_2 \rightarrow P(s_2, v) \rightarrow v, (v)^n \rightarrow P((v)^n, t_2) \rightarrow t_2\rangle$ are two spanning disjoint paths of $S_n^I - F_{av}$, as illustrated in Fig. 3.

Case 2 $\{s_1, s_2\}, \{s_1, t_2\}, \{t_1, s_2\}, \{t_1, t_2\} \not\subset S_n^m$ for every $m \in I$.

Without loss of generality, we can assume that $s_1 \in S_n^{i_1}$, $t_1 \in S_n^{i_j}$, $s_2 \in S_n^{i_m}$, and $t_2 \in S_n^{i_k}$ for $j < m$. Let $J_1 = \{i_1, \cdots, i_j\}$ and $J_2 = \{i_{j+1}, \cdots, i_k\}$. By Lemma 4, there exists a Hamiltonian path $P(s_1, t_1)$ of $S_n^{J_1} - F_{av}$ and a Hamiltonian path $P(s_2, t_2)$ of $S_n^{J_2} - F_{av}$. Thus, $P(s_1, t_1)$ and $P(s_2, t_2)$ are two spanning disjoint paths of $S_n^I - F_{av}$, as illustrated in Fig. 4.

Case 3 exactly one of $\{s_1, s_2\}, \{s_1, t_2\}, \{t_1, s_2\}, \{t_1, t_2\}$ in S_n^m for some $m \in I$.

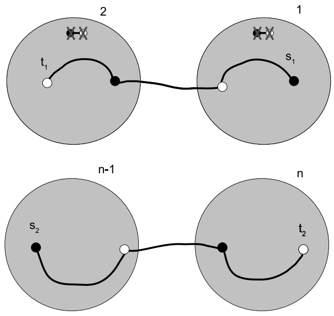

Figure 4: The case 2 for two spanning disjoint paths of Lemma 5.

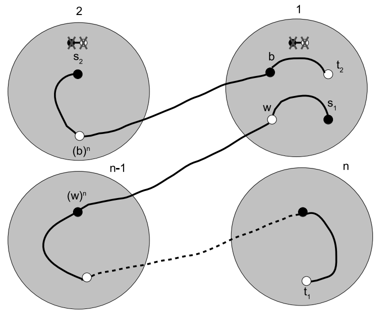

Figure 5: The case 3.2 for two spanning disjoint paths of Lemma 5.

Case 3.1 $\{s_1, s_2\}$ or $\{t_1, t_2\}$ in S_n^k for some $k \in I$.
Without loss of generality, we can assume that $s_1, s_2, \in S_n^{i_1}$, $t_1 \in S_n^{i_2}$ and $t_2 \in S_n^{i_k}$. Let $w_1, w_2 \in V_1$ be two vertices of $S_n^{i_1} - F_{av}$ such that $(w_1)^n \in S_n^{i_2}, (w_2)^n \in S_n^{i_3}$. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, there exist two spanning disjoint paths $P(s_1, w_1)$ and $P(s_2, w_2)$ of $S_n^{i_1} - F_{av}$. Let $J = I - \{i_1, i_2\}$. By Lemma 4, there exists a Hamiltonian path $P((w_1)^n, t_1)$ of $S_n^{i_2} - F_{av}$ and a Hamiltonian path $P((w_2)^n, t_2)$ of $S_n^J - F_{av}$. Thus, the two paths $\langle s_1 \to P(s_1, w_1) \to w_1, (w_1)^n \to P((w_1)^n, t_1) \to t_1\rangle$ and $\langle s_2 \to P(s_2, w_2) \to w_2, (w_2)^n \to P((w_2)^n, t_2) \to t_2\rangle$ form two spanning disjoint paths of $S_n^I - F_{av}$.

Case 3.2 $\{s_1, t_2\}$ or $\{t_1, s_2\}$ in S_n^m for some $m \in I$.
Without loss of generality, we can assume that $s_1, t_2, \in S_n^{i_1}$, $t_1 \in S_n^{i_2}$ and $s_2 \in S_n^{i_k}$. Let $w \in V_1, b \in V_0$ be two vertices of $S_n^{i_1} - F_{av}$ such that $(w)^n \in S_n^{i_2}, (b)^n \in S_n^{i_3}$. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, there exist two spanning disjoint paths $P(s_1, w)$ and $P(b, t_2)$ of $S_n^{i_1} - F_{av}$. Let $J = I - \{i_1, i_2\}$. By Lemma 4, there exists a Hamiltonian path $P((w)^n, t_1)$ of $S_n^{i_2} - F_{av}$ and a Hamiltonian path $P(s_2, (b)^n)$ of $S_n^J - F_{av}$. Thus, the two paths $\langle s_1 \to P(s_1, w) \to w, (w)^n \to P((w)^n, t_1) \to t_1\rangle$ and $\langle s_2 \to P(s_2, (b)^n) \to (b)^n, b \to P(b, t_2) \to t_2\rangle$ form two spanning disjoint paths of $S_n^I - F_{av}$, as illustrated in Fig. 5.

Case 4 two of $\{s_1, s_2\}, \{s_1, t_2\}, \{t_1, s_2\}, \{t_1, t_2\}$ in S_n^m for some $m \in I$.

Case 4.1 $\{s_1, s_2\}$ in S_n^i and $\{t_1, t_2\}$ in S_n^j for $i \neq j \in I$.
Without loss of generality, we can assume that $\{s_1, s_2\}$ in $S_n^{i_1}$ and $\{t_1, t_2\}$ in $S_n^{i_k}$. By Lemma 1, there exists a Hamiltonian path $P((w_2)^n, (b)^n)$ of $S_n^J - F_{av}$.

Let $w_1, w_2 \in V_1$ be two vertices of $S_n^{i_1} - F_{av}$ such that $(w_1)^n \in S_n^{i_k}, (w_2)^n \in S_n^{i_2}$. Let $b \in V_0$ be a vertex of $S_n^{i_k} - F_{av}$ for $(b)^n \in S_n^{i_{k-1}} - F_{av}$. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, there exist two spanning disjoint paths $P(s_1, w_1)$ and $P(s_2, w_2)$ of $S_n^{i_1} - F_{av}$ and two spanning disjoint paths $P((w_1)^n, t_1)$ and $P(b, t_2)$ of $S_n^{i_k} - F_{av}$. Let $J = I - \{i_1, i_k\}$. By Lemma 1, there exists a Hamiltonian path $P((w_2)^n, (b)^n)$ of $S_n^J - F_{av}$. Thus, the two paths $\langle s_1 \to P(s_1, w_1) \to w_1, (w_1)^n \to P((w_1)^n, t_1) \to t_1\rangle$ and $\langle s_2 \to P(s_2, w_2) \to w_2, (w_2)^n \to P((w_2)^n, (b)^n) \to (b)^n, b \to P(b, t_2) \to t_2\rangle$ form two spanning disjoint paths of $S_n^I - F_{av}$.

Case 4.2 $\{s_1, t_2\}$ in S_n^i and $\{t_1, s_2\}$ in S_n^j for $i \neq j \in I$.
Without loss of generality, we can assume that $\{s_1, t_2\}$ in $S_n^{i_1}$ and $\{t_1, s_2\}$ in $S_n^{i_k}$. Let $w \in V_1$ and $b \in V_0$ be two vertices of $S_n^{i_1} - F_{av}$ such that $(w)^n \in S_n^{i_k} - F_{av}, (b)^n \in S_n^{i_2} - F_{av}$. Let $c \in V_1$ be a vertex of $S_n^{i_k} - F_{av}$ for $(c)^n \in S_n^{i_{k-1}} - F_{av}$. Since S_{n-1} is $(n-5)$-adjacent 2-spannable, there exist two spanning disjoint paths $P(s_1, w)$ and $P(b, t_2)$ of $S_n^{i_1} - F_{av}$ and two spanning disjoint paths $P((w)^n, t_1)$ and $P(s_2, c)$ of $S_n^{i_k} - F_{av}$. Let $J = I - \{i_1, i_k\}$. By Lemma 4, there exists a Hamiltonian path $P((c)^n, (b)^n)$ of $S_n^J - F_{av}$. Thus, the two paths $\langle s_1 \to P(s_1, w) \to w, (w)^n \to P((w)^n, t_1) \to t_1\rangle$ and $\langle s_2 \to P(s_2, c) \to c, (c)^n \to P((c)^n, (b)^n) \to (b)^n, b \to P(b, t_2) \to t_2\rangle$ form two spanning disjoint paths of $S_n^I - F_{av}$, as illustrated in Fig. 6. $\square$

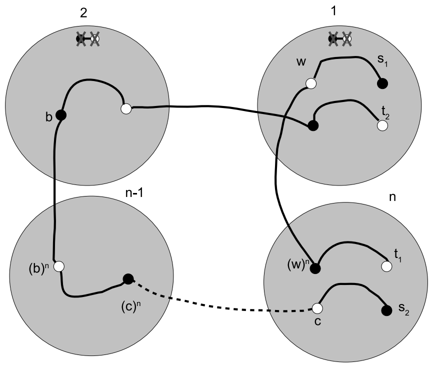

Figure 6: The case 4.2 for two spanning disjoint paths of Lemma 5.

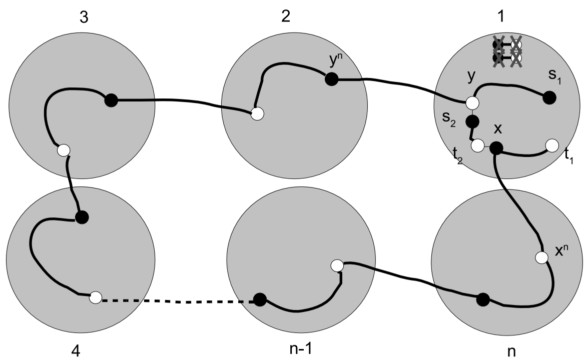

Figure 7: The case 1.1 for two spanning disjoint paths of Theorem 1.

Theorem 1 *The graph S_n is $(n-4)$-adjacent 2-spannable for $n \geq 5$.*

Proof. We will prove this theorem by induction on n. The case for $n = 5$ holds in Lemma 3. In the follows, we will assume that $n \geq 6$. By induction hypothesis, we can assume that S_{n-1} is $(n-5)$-adjacent 2-spannable. Let F_{av} be the set of faulty vertices with f_{av} adjacent pairs of vertices in S_n. Let f^i_{av} be the number of pairs of faulty vertices in S^i_n for $i \in \langle n \rangle$. By the symmetry of star graphs, we can assume that every pair of F_{av} is in the same subgraph S^i_n for $i \in \langle n \rangle$. Let $s_1, s_2 \in V_0$ and $t_1, t_2 \in V_1$ be arbitrary two pairs of $S_n - F_{av}$. Let $U = \{s_1, s_2, t_1, t_2\}$ and $U^i = U \cap V(S^i_n)$. We will construct two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of $S_n - F_{av}$ for $f_{av} \leq n-4$ in the following cases.

Case 1 $f^i_{av} = n-4$ for some $i \in \langle n \rangle$.
Without loss of generality, we can assume that $f^1_{av} = n-4$.

Case 1.1 $|U^1| = 4$.
By Lemma 2, $S^1_n - F_{av}$ is Hamiltonian laceable. There exists a Hamiltonian path $P(s_1, t_1)$. Without loss of generality, we can denote $P(s_1, t_1)$ as $\langle s_1 \to P(s_1, y) \to y, s_2 \to P(s_2, t_2) \to t_2, x \to P(x, t_1) \to t_1 \rangle$. Let $J = \langle n \rangle - \{1\}$. By Lemma 4, there exists a Hamiltonian path $P((y)^n, (x)^n)$ of S^J_n. Thus, $P(s_2, t_2)$ and $\langle s_1 \to P(s_1, y) \to y, (y)^n \to P((y)^n, (x)^n) \to (x)^n, x \to P(x, t_1) \to t_1 \rangle$ are two spanning disjoint paths of $S_n - F_{av}$, as illustrated in Fig. 7.

Case 1.2 $|U^1| = 3$.
Without loss of generality, we can assume that $U^1 = \{s_1, t_1, t_2\}$. By Lemma 2, S^1_n is Hamiltonian laceable. There exists a Hamiltonian path $P(s_1, t_2)$ of $S^1_n - F_{av}$. We can denote $P(s_1, t_2)$ as $\langle s_1 \to P(s_1, t_1) \to t_1, x \to P(x, t_2) \to t_2 \rangle$. Let $J = \langle n \rangle - \{1\}$. By Lemma 4, there exists a Hamiltonian path $P(s_2, (x)^n)$ of S^J_n. Thus, $P(s_1, t_1)$ and $\langle s_2 \to P(s_2, (x)^n) \to (x)^n, x \to P(x, t_2) \to t_2 \rangle$ are two spanning disjoint paths of $S_n - F_{av}$.

Case 1.3 $|U^1| = 2$.

Case 1.3.1 $U^1 = \{s_1, t_1\}$ or $U^1 = \{s_2, t_2\}$.
Without loss of generality, we can assume that $U^1 = \{s_1, t_1\}$. By Lemma 2, S^1_n is Hamiltonian laceable. There exists a Hamiltonian path $P(s_1, t_1)$ of $S^1_n - F_{av}$. Let $J = \langle n \rangle - \{1\}$. By Lemma 4, there exists a Hamiltonian path $P(s_2, t_2)$ of S^J_n. Thus, $P(s_1, t_1)$ and $P(s_2, t_2)$ are two spanning disjoint paths of $S_n - F_{av}$.

Case 1.3.2 $U^1 = \{s_1, t_2\}$ or $U^1 = \{s_2, t_1\}$.
Without loss of generality, we can assume that $U^1 = \{s_1, t_2\}$. By Lemma 2, there exists a Hamiltonian path $P(s_1, t_2)$ of $S^1_n - F_{av}$. Since the length of $P(s_1, t_2) \geq 7$, we can denote $P(s_1, t_2)$ as $\langle s_1 \to P(s_1, y) \to y, x \to P(x, t_2) \to t_2 \rangle$ with $y \in V_1, x \in V_0$ and $\{s_2, t_1\} \cap \{(x)^n, (y)^n\} = \emptyset$. Let $J = \langle n \rangle - \{1\}$. By Lemma 5, there exist two spanning disjoint paths $P(s_2, (x)^n)$ and $P((y)^n, t_1)$. Thus, $\langle s_1 \to P(s_1, y) \to y, (y)^n \to P((y)^n, t_1) \to t_1 \rangle$ and $\langle s_2 \to P(s_2, (x)^n) \to (x)^n, x \to P(x, t_2) \to t_2 \rangle$ are two spanning disjoint paths of

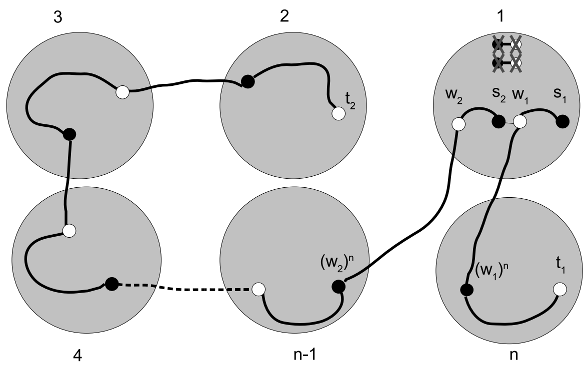

Figure 8: The case 1.3.3 for two spanning disjoint paths of Theorem 1.

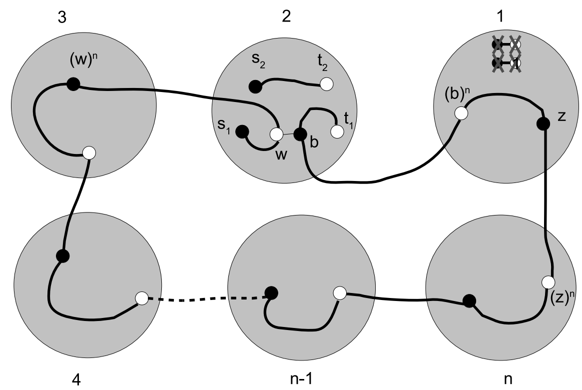

Figure 9: The case 1.5.3 for two spanning disjoint paths of Theorem 1.

$S_n - F_{av}$.

Case 1.3.3 $U^1 = \{s_1, s_2\}$ or $U^1 = \{t_1, t_2\}$.
Without loss of generality, we can assume that $U^1 = \{s_1, s_2\}$. Let w_2 be a vertex of $V(S_n^1) \cap V_1$. By Lemma 2, there exists a Hamiltonian path $P(s_1, w_2)$. We can denote the path $P(s_1, w_2)$ as $\langle s_1 \to P(s_1, w_1) \to w_1, s_2 \to P(s_2, w_2) \to w_2\rangle$. Let $J = \langle n\rangle - \{1\}$. By Lemma 5, there exist two spanning disjoint paths $P((w_1)^n, t_1)$ and $P((w_2)^n, t_2)$ of S_n^J. Thus, $\langle s_1 \to P(s_1, w_1) \to w_1, (w_1)^n \to P((w_1)^n, t_1) \to t_1\rangle$ and $\langle s_2 \to P(s_2, w_2) \to w_2, (w_2)^n \to P((w_2)^n, t_2) \to t_2\rangle$ are two spanning disjoint paths of $S_n - F_{av}$, as illustrated in Fig. 8.

Case 1.4 $|U^1| = 1$.
Without loss of generality, we can assume that $U^1 = \{s_1\}$. Let $w_1 \in V_1$ be a vertex of $S_n^1 - F_{av}$ with $(w_1)^n \neq s_2$. By Lemma 2, there exists a Hamiltonian path $P(s_1, w_1)$ of $S_n^1 - F_{av}$. Let $J = \langle n\rangle - \{1\}$. By Lemma 5, there exist two spanning disjoint paths $P((w_1)^n, t_1)$ and $P(s_2, t_2)$ of S_n^J. Thus, $\langle s_1 \to P(s_1, w_1) \to w_1, (w_1)^n \to P((w_1)^n, t_1) \to t_1\rangle$ and $P(s_2, t_2)$ are two spanning disjoint paths of $S_n - F_{av}$.

Case 1.5 $|U^1| = 0$.

Case 1.5.1 $|U^i| = 1$ for some $2 \leq i \leq n$.
Without loss of generality, we can assume that $U^2 = \{s_2\}$. Let $w_1 \in V(S_n^1)$ and $w_2 \in V(S_n^2)$ be two vertices of V_1 with $(w_1)^n \neq s_1$ and $(w_2)^n \in V(S_n^1) - F_{av}$. By Lemma 2, there exists a Hamiltonian path $P(s_2, w_2)$ of S_n^2 and a Hamiltonian path $P((w_2)^n, w_1)$ of $S_n^1 - F_{av}$. Let $J = \langle n\rangle - \{1, 2\}$. By Lemma 5, there exist two spanning disjoint paths $P(s_1, t_1)$ and $P((w_1)^n, t_2)$ of S_n^J. Thus, $P(s_1, t_1)$ and $\langle s_2 \to P(s_2, w_2) \to w_2, (w_2)^n \to P((w_2)^n, w_1) \to w_1, (w_1)^n \to P((w_1)^n, t_2) \to t_2\rangle$ are two spanning disjoint paths of $S_n - F_{av}$.

Case 1.5.2 $|U^i| = 2$ for some $2 \leq i \leq n$.
Without loss of generality, we can assume that $U^n = \{s_2, t_2\}$. Let $w \in V_1$, $x \in V_0$ be two vertices of S_n^n with $(w)^n \neq s_1, (w)^n \notin V(S_n^1)$ and $(x)^n \in V(S_n^1) - F_{av}$. Let $z \in V_0$ be a vertex of $S_n^1 - F_{av}$ for $(z)^n \notin V(S_n^n)$ and $(z)^n \neq t_1$. By Lemma 2, there exists a Hamiltonian path $P(z, (x)^n)$ of $S_n^1 - F_{av}$. By Lemma 5, there exist two spanning disjoint paths $P(x, t_2)$ and $P(s_2, w)$ of S_n^n. Let $J = \langle n\rangle - \{1, n\}$. By Lemma 5, there exist two spanning disjoint paths $P(s_1, t_1)$ and $P((w)^n, (z)^n)$ of S_n^J. Thus, $P(s_1, t_1)$ and $\langle s_2 \to P(s_2, w) \to w, (w)^n \to P((w)^n, (z)^n) \to (z)^n, z \to P(z, (x)^n) \to (x)^n, x \to P(x, t_2) \to t_2\rangle$ are two spanning disjoint paths of $S_n - F_{av}$.

Case 1.5.3 $|U^i| = 4$ for some $2 \leq i \leq n$.
By Lemma 5, there exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of S_n^i. Without loss of generality, we can denote the paths $P(s_1, t_1)$ as $\langle s_1 \to P(s_1, w) \to w, b \to P(b, t_1) \to t_1\rangle$ with $(b)^n \in V(S_n^1) - F_{av}$. Let $z \in V_0$ be a

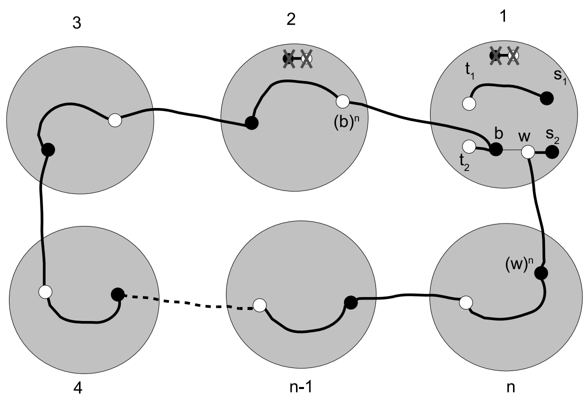

Figure 10: The case 2.1 for two spanning disjoint paths of Theorem 1.

vertices of $S_n^1 - F_{av}$ for $(z)^n \notin V(S_n^i)$. By Lemma 2, there exists a Hamiltonian path $P(z,(b)^n)$ of $S_n^1 - F_{av}$. Let $J = \langle n \rangle - \{1, i\}$. By Lemma 4, there exists a Hamiltonian path $P((w)^n,(z)^n)$ of S_n^J. Thus, $\langle s_1 \to P(s_1,w) \to w,(w)^n \to P((w)^n,(z)^n) \to (z)^n, z \to P(z,(b)^n) \to (b)^n, b \to P(b,t_1) \to t_1\rangle$ and $P(s_2,t_2)$ are two spanning disjoint paths of $S_n - F_{av}$, as illustrated in Fig. 9.

Case 2 $f_{av}^i \le n-5$ for every $i \in \langle n \rangle$.

Case 2.1 $|U^i| = 4$ for some $1 \le i \le n$.
By induction hypothesis, there exist two spanning disjoint paths $P(s_1,t_1)$ and $P(s_2,t_2)$ of $S_n^i - F_{av}$. Without loss of generality, we can denote the path $P(s_2,t_2)$ as $\langle s_2 \to P(s_2,w) \to w, b \to P(b,t_2) \to t_2\rangle$ of $S_n^i - F_{av}$ with $(b)^n,(w)^n \notin F_{av}$. Let $J = \langle n \rangle - \{i\}$. By Lemma 4, there exists a Hamiltonian path $P((w)^n,(b)^n)$ of $S_n^J - F_{av}$. Thus, $P(s_1,t_1)$ and $\langle s_2 \to P(s_2,w) \to w,(w)^n \to P((w)^n,(b)^n) \to (b)^n, b \to P(b,t_2) \to t_2\rangle$ are two spanning disjoint paths of $S_n - F_{av}$, as illustrated in Fig. 10.

Case 2.2 $|U^i| \le 3$ for every $1 \le i \le n$.
By Lemma 5, there exist two spanning disjoint paths $P(s_1,t_1)$ and $P(s_2,t_2)$ of $S_n - F_{av}$. $\square$

4 Conclusion

We have showed that S_n is $(n-4)$-adjacent 2-spannable for $n \ge 5$. It is worth to investigate the edges fault-tolerance for this property.

References

[1] Chun-Nan Hung, Yi-Hua Chang, and Chao-Min Sun, "Longest paths and cycles in fault hypercubes," *Proceedings of the IASTED International Conference on Parallel and Distributed Computing and Networks*, (2006) pp. 101-110.

[2] Chun-Nan Hung, Chi-Lai Liu and Hsuan-Han Chang, "Edge Fault Tolerance for Two-Pair Spanning Disjoint Paths," *Proceedings of the 25th Workshop on Combinatorial Mathematics and Computation Theory*, (2008) pp. 375-384.

[3] Tseng-Kuei Li,Jimmy J.M. Tan, Lih-Hsing Hsu, "Hyper hamiltonian laceability on edge fault star graph," *Information Sciences*, 165 (2004) pp. 59-71.

[4] Tsung-Han Ou and Chun-Nan Hung, "The Adjacent Vertices Fault Tolerance Hamiltonian Laceability of Star Graphs," *Proceedings of the 2011 National Computer Symposium Workshop on Algorithms and Bioinformatics*, (2011) pp. 65-71.

[5] Wen-Yan Su and Chun-Nan Hung, 4, "The Longest Ring Embedding in Faulty Hypercube," *Proceedings of the 23rd Workshop on Combinatorial Mathematics and Computational Theory*, (2006) pp. 262-272.

[6] Chang-Hsiung Tsai, Jimmy J.M. Tan, Tyne Liang, and Lih-Hsing Hsu, "Fault-tolerant hamiltonian laceability of hypercubes," *Information Processing Letters*, 83 (2002) pp. 301-306.